

Mathematics A

Unit 2 • Chapter OL-T38 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

7 classified items • 34 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T38 • Expertise Q16+ • 7 items • 34 marks

Chapter	Items	Marks	Starts
01. OL-T38 Differentiation	7	34	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Differentiation

Unit 2 • Expertise • 34 marks • 7 item(s)

June 2025 | 4MA1-2H Q17 | 6 marks

Coordinates of polynomial stationary points

November 2023 | 4MA1-2H Q18 | 4 marks

Velocity and acceleration from displacement

November 2024 | 4MA1-1H Q20 | 4 marks

Equation of a calculus tangent

January 2022 | 4MA1-1H Q24 | 6 marks

Recover curve parameters from point and gradient data

January 2022 | 4MA1-2H Q24 | 5 marks

Instantaneous rest or direction reversal

May-Nov 2020 | 1H Q23 | 4 marks

Points where gradient equals a value

January 2023 | 1H Q20 | 5 marks

Maximum volume from geometric dimensions

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

$$\dots\dots\dots (4)$$

(Total for Question 17 is 6 marks)

WORKING SPACE

Derivative and turning points

6 marks

Differentiate

$$\frac{dy}{dx} = 12x^2 + 10x + 2.$$

Solve derivative equal to zero

$$12x^2 + 10x + 2 = 2(3x + 1)(2x + 1) = 0 \Rightarrow x = -\frac{1}{3}, -\frac{1}{2}.$$

Substitute

$$x = -\frac{1}{2} \Rightarrow y = -\frac{1}{4}; \quad x = -\frac{1}{3} \Rightarrow y = -\frac{7}{27}.$$

Final answer

$$\frac{dy}{dx} = 12x^2 + 10x + 2; \quad \left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$$

4MA1-2H Q18 whole

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q18 whole

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

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Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

Acceleration from a displacement function

4 marks

whole – Acceleration from a displacement function

Differentiate once

$$v = \frac{ds}{dt} = 6t^2 - 10t + 6.$$

Differentiate again

$$a = \frac{dv}{dt} = 12t - 10.$$

Set the required acceleration

$$12t - 10 = 5 \Rightarrow 12t = 15.$$

Solve

$$t = \frac{15}{12} = 1.25 \text{ s.}$$

Final answer

$$t = 1.25 \text{ s}$$

4MA1-1H Q20 M01

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

.....

WORKING SPACE

Find the tangent at a minimum

4 marks

M01 – Find the tangent at a minimum**Differentiate**

$$\frac{dy}{dx} = 8x^3 - 64.$$

Set the gradient to zero

$$8x^3 - 64 = 0 \Rightarrow x = 2.$$

Find the y-coordinate

$$y = 2(2^4) - 64(2) = -96.$$

State the horizontal tangent

$$y = -96.$$

Final answer

$$y = -96$$

4MA1-1H Q24

6 marks

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q24

6 marks

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

Chapter: Differentiation

6 marks

Q24 – Chapter: Differentiation

Family: Find and classify stationary points • Variant: Recover curve parameters from point and gradient data

Method

At $(2, -6)$, $-6 = 8a + 4b - 24 + 6$, so $8a + 4b = 12$.

Why: Substitute the point into the cubic equation.

Method

$dy/dx = 3ax^2 + 2bx - 12$.

Why: Differentiate the curve.

Method

At $x=2$, $16 = 12a + 4b - 12$, so $12a + 4b = 28$.

Why: Use the given gradient at A.

Method

Solving $8a + 4b = 12$ and $12a + 4b = 28$ gives $a = 4$ and $b = -5$.

Why: Solve the simultaneous linear equations.

Method

At $x=3$, $y = 4(3)^3 - 5(3)^2 - 12(3) + 6$.

Why: Substitute $x=3$ and the constants.

Method

$y = 33$.

Why: Evaluate the expression.

Final answer

y: 33

4MA1-2H Q24

5 marks

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q24

5 marks

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

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The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

Chapter: Differentiation

5 marks

Q24 – Chapter: Differentiation

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

Method

$$v = dx/dt = 12t^2 - 27.$$

Why: Differentiate $x = 4t^3 - 27t + 8$.

Method

At a reversal $v=0$, so $12t^2 - 27 = 0$ and $t^2 = 9/4$.

Why: Set velocity equal to zero.

Method

Since $t \geq 0$, $t = 3/2$.

Why: Choose the physical time.

Method

$$a = dv/dt = 24t.$$

Why: Differentiate velocity.

Method

At $t = 3/2$, $a = 36 \text{ m/s}^2$.

Why: Evaluate the acceleration.

Final answer

acceleration: 36 m/s^2

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

Answer reference | May-Nov 2020 | Question 23

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

23	$\left(\frac{dy}{dx} = 3px^2 - m\right)$		4	M1 for $3px^2$ or $-m$
	$3px^2 - m < 0$ oe			M1 ft dep on M1 for setting up an inequality with their ' $3px^2$ ' - ' m ' must be a two-term expression in the form $apx^2 \pm m$
	$\pm \sqrt{\frac{m}{3p}}$			B1 for both critical values
		$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$		A1 may be seen as two separate inequalities
Total 4 marks				

Worked solution

May-Nov 2020 | Question 23 | 4 marks

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

Method

$$y = px^3 - mx$$

Differentiate:

$$(dy)/(dx) = 3px^2 - m$$

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < (m)/(3p)$$

So

$$-\sqrt{(m)/(3p)} < x < \sqrt{(m)/(3p)}$$

$$\text{Answer: } -\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is V cm³

Find the maximum value of V

Give your answer correct to the nearest whole number.



(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

20	$(V =) \pi x^2 \left(\frac{800}{\pi x} - x \right)$ or $800x - \pi x^3$ oe		5	M1 for volume of cylinder (in terms of one variable, e.g. x or r)
	$\left(\frac{dV}{dx} = \right) 800 - 3\pi x^2$			M1ft for differentiating an expression in one variable to find 800 or $\pm 3\pi x^2$ (must come from a cubic in the form $800x \pm \pi x^3$ or $800x \pm ax^3$ or $bx \pm \pi x^3$ where $a \neq 0$ and $b \neq 0$)
	" $800 - 3\pi x^2 = 0$ "			M1ft dep on previous M1 for equating their $\frac{dV}{dx}$ to zero (must be a quadratic in the form $800 \pm ax^2$ or $b \pm 3\pi x^2$ where $a \neq 0$ and $b \neq 0$)
	$(x =) \sqrt{\frac{800}{3\pi}} (= \sqrt{84.8(8263...)})$ or $\frac{\sqrt{9600\pi}}{6\pi}$ or $9.2(13177...)$			A1 for a correct value of x Allow use of quadratic formula
	Award marks within the range from correct working	4914		A1 accept 4910 – 4914
				Total 5 marks

Worked solution

January 2023 | Question 20 | 5 marks

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

Official final mark-scheme pages are embedded immediately after this question.